Machine Learning in Business

MIS710 – A2

Predicting Writing Underperformance in Primary School Students: A Data-Driven Approach to Early Interventions

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Executive Summary (Max 100 Words)

This report explores the application of machine learning to predict students at risk of underperforming in writing assessments. Using exploratory data analysis and predictive modeling, insights were generated to answer key business questions for Data2Intel. The Gradient Boosting model was chosen for its superior performance in predicting at-risk students due to its ability to capture complex data relationships. The report includes recommendations for business applications, resource allocation, and process improvements to enhance educational outcomes and decision-making in schools.

1. Introduction

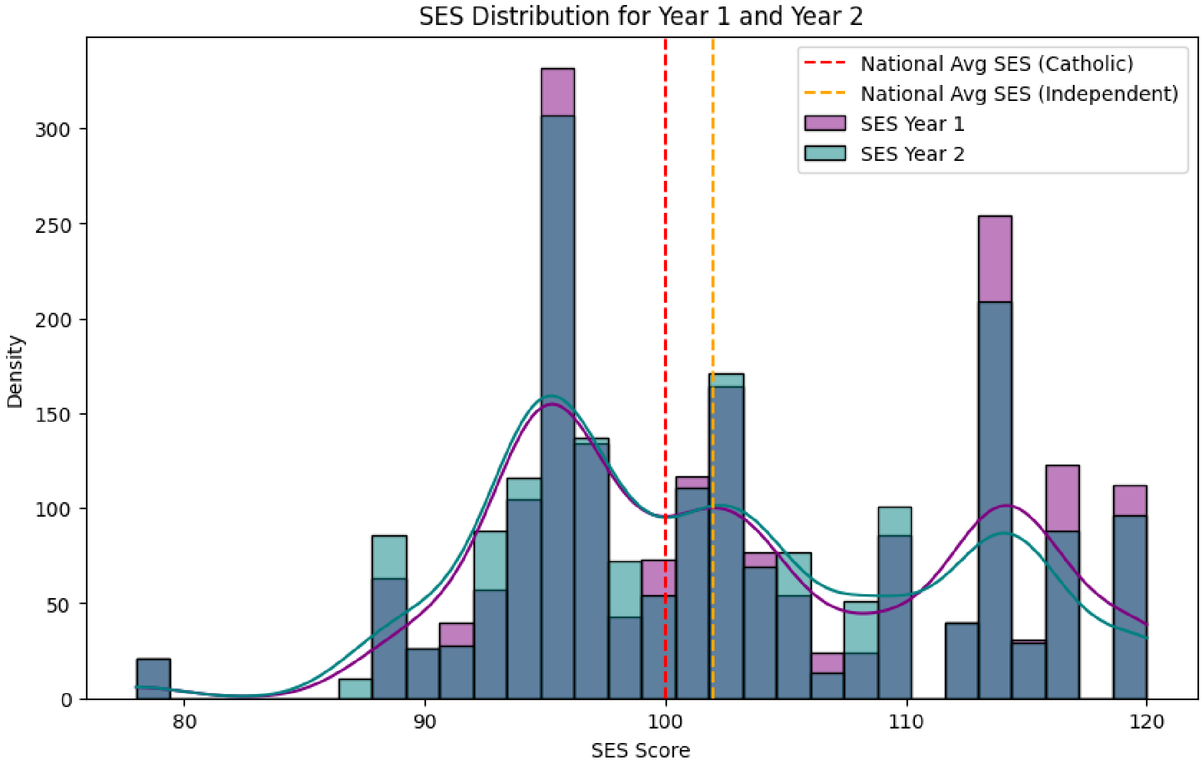
1.1 Business Understanding (BACCM Framework)

The **Business Analysis Core Concept Model (BACCM)** framework provides a structured approach to understanding the business context of this project. Each core concept within BACCM helps identify how machine learning will add value to the education system:

- **Change**: The predictive model aims to drive early interventions for students at risk, allowing schools to allocate resources efficiently and improve outcomes.
- **Need**: Schools need an accurate system to identify students struggling with writing skills in the early years to provide timely support.
- **Solution**: The machine learning model provides actionable insights for targeting interventions based on historical academic performance data.
- **Stakeholders**: Stakeholders include teachers, administrators, parents, and students. Each benefits from improved educational planning and resource allocation.
- **Value**: The model helps increase the effectiveness of educational interventions by focusing on students who need them the most, ultimately enhancing student outcomes.
- **Context**: The context is Australia's primary school system, where data-driven interventions are becoming critical for improving literacy and numeracy outcomes.

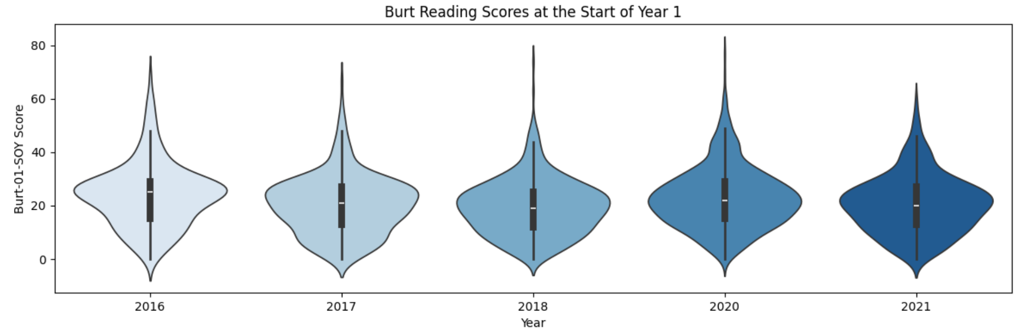
2. Insights from Exploratory Data Analysis (EDA)
2.1 Answers to Client's Six Questions

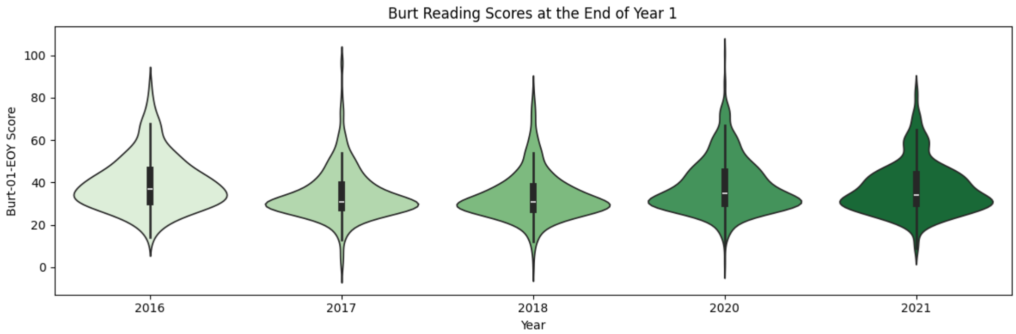
1. **SES Distribution:** Students' SES scores vary widely, with many falling below the national averages of 100 (Catholic) and 102 (Independent) in 2018. A significant portion of students have SES scores between 90 and 100, particularly in Year 1. This highlights the need for early interventions for students from lower SES backgrounds, which likely correlates with their academic performance.



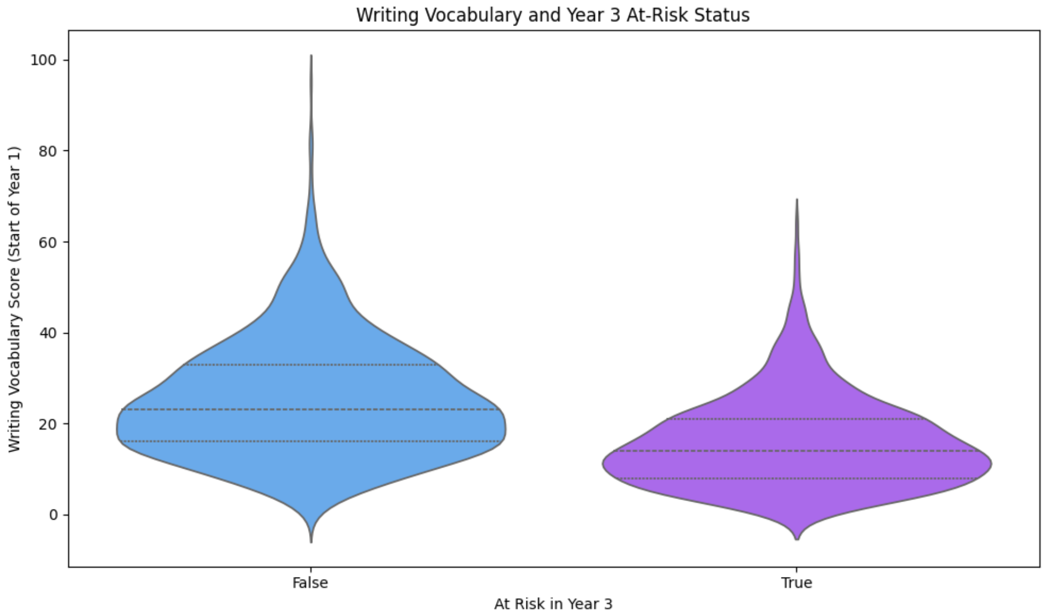
2. **Burt Reading Scores:** Burt Reading Scores improve from the start to the end of Year 1 across the years (2016–2021). While students generally improve their reading skills during the year, the score distribution shows variability, with some students making significant progress while others continue to struggle. These

students require additional support.

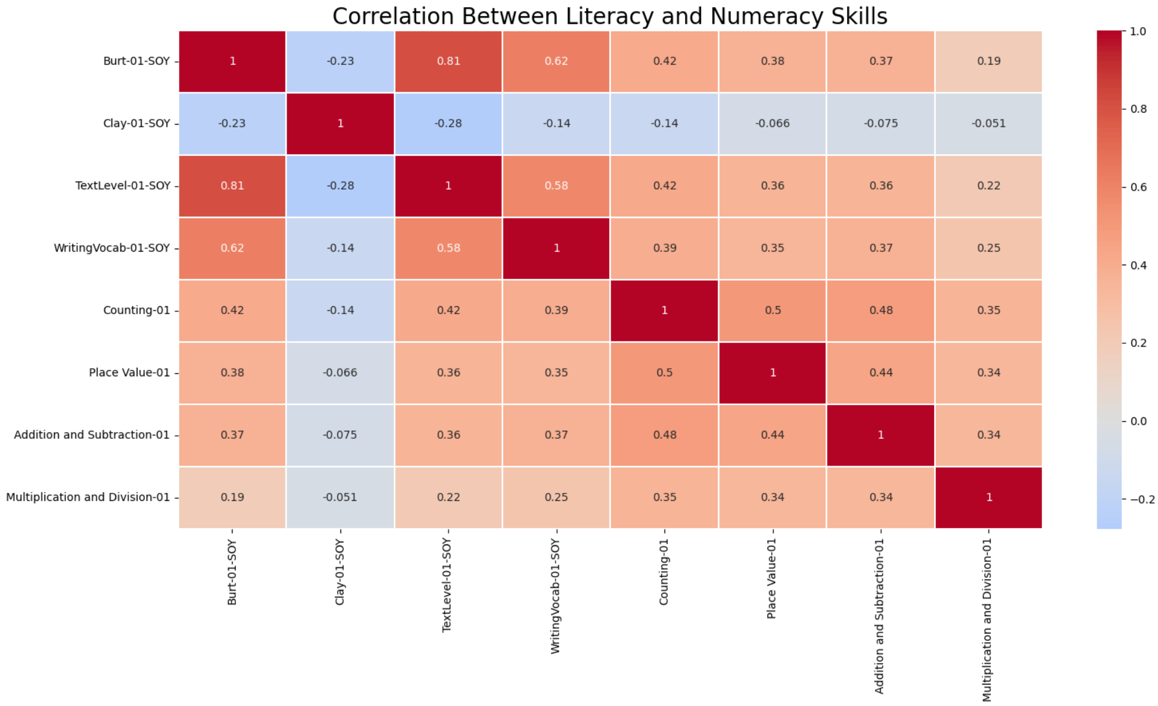




3. **Writing Vocabulary** and At-Risk Status: Students not at risk in Year 3 had higher Writing Vocabulary scores at the start of Year 1, with a median score around 40. Those classified as "at risk" had significantly lower scores, with a median of about 20. This suggests that low early writing skills are a predictor of future underperformance, reinforcing the need for early interventions.

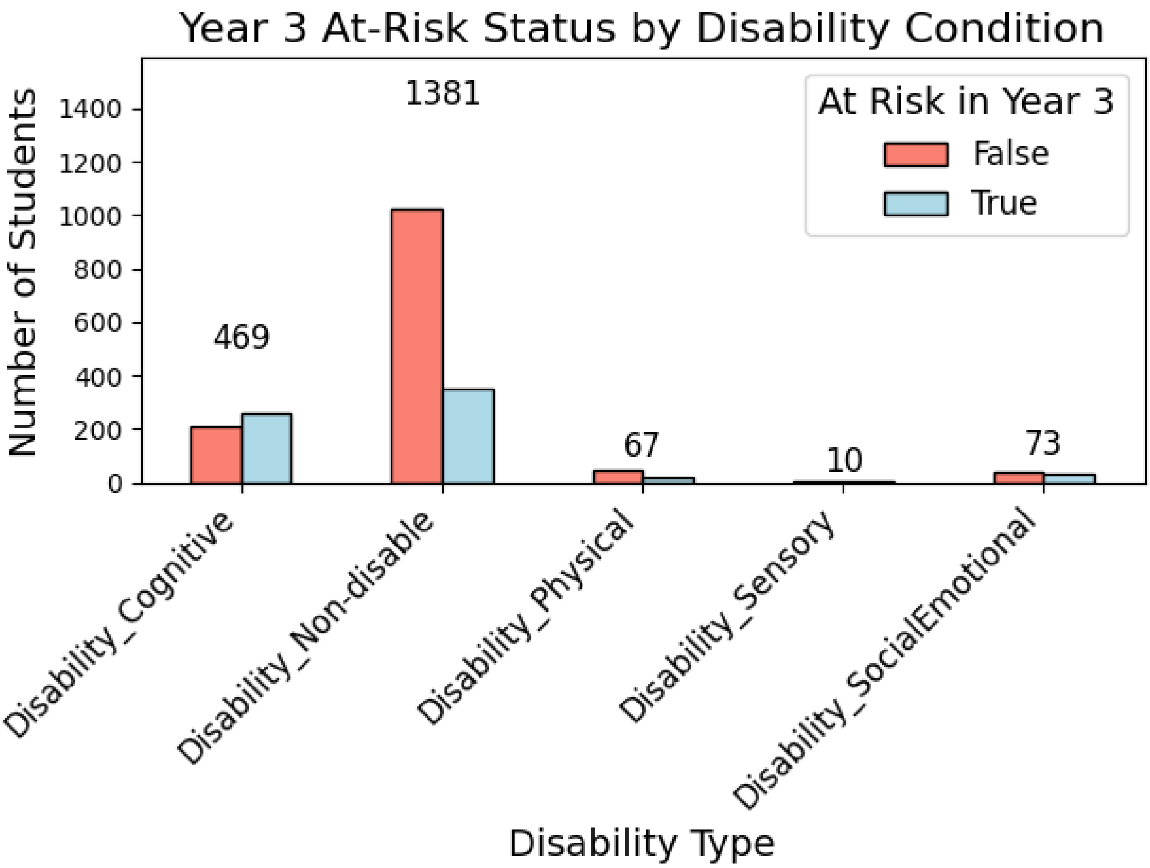


4. **Literacy and Numeracy Relationships**: The heatmap shows strong correlations between literacy and numeracy skills. Burt Reading Scores and Text Level are highly correlated (0.81), as are Writing Vocabulary and Burt Reading (0.62). Students who perform well in literacy also tend to excel in numeracy tasks like Counting and Place Value. Poor performance in these areas increases the likelihood of being classified as at risk.



5. **Disability Conditions** and At-Risk Status: Most students do not have a disability, but among those who do, students with Cognitive and Social-Emotional Disabilities are disproportionately at risk of underperforming in Year 3 writing. These conditions likely impact their learning process, indicating the need for

targeted interventions.



6. **Clustering Insights:** Students with moderate literacy and numeracy skills but lower SES are ideal candidates for early intervention. Targeted programs that focus on both skill areas can significantly improve their writing performance by Year 3.

3. Proposed Machine Learning Solution

3.1 Selected Model: Gradient Boosting Classifier

The **Gradient Boosting Classifier** was selected for its high accuracy (73.67%) and superior ROC AUC (0.7511), outperforming the Random Forest model. It effectively captures complex patterns, particularly in identifying students at risk of underperformance.

3.2 Interpretation of Performance

The model accurately identifies at-risk students by focusing on key predictors such as literacy scores and SES. It iteratively improves predictions, making it highly effective, though computationally intensive and less interpretable for non-technical stakeholders.

Pros:

- High accuracy in identifying at-risk students.
- Strong performance on complex data relationships.

Cons:

- Computationally intensive.
- Less interpretable compared to simpler models like Random Forest.

4. Recommendations and Conclusions

4.1 Business Applications

- **Early Intervention Programs**: The model enables schools to implement targeted interventions for students at risk, such as tutoring, specialized programs, or increased family engagement.
- **Resource Allocation**: Schools can use the model to allocate resources efficiently, ensuring maximum impact on students identified as needing the most support.

4.2 Benefits to Stakeholders

- **Teachers**: Gain insights into students needing the most attention and tailor teaching strategies.
- **School Administrators**: Make data-driven decisions on resource allocation and program funding.
- **Parents**: Receive early warnings about their child's risk, allowing them to take proactive steps.
- **Students**: Benefit from personalized support, improving academic performance and long-term outcomes.

4.3 Impacts on Business Processes and Decision Making

- **Data-Driven Decision Making**: Schools will integrate the model into decision-making processes for early identification of at-risk students.
- **Process Changes**: Implementing the model may require adjustments in data collection and storage. Regular updates are essential to maintain accuracy.
- **Impact on Teaching**: Teachers may need training to understand and use model outputs effectively for classroom interventions.

4.4 Recommendations for Further Improvements

- **Incorporate Additional Data**: Adding data such as attendance records and behavioural metrics could improve the model's accuracy.
- **Model Explainability**: Developing tools to make the Gradient Boosting model more interpretable will help stakeholders better understand the results.

- **Regular Model Updates**: The model should be retrained regularly with new data to ensure continued relevance and predictive power.